Price Policy as a Driver of Agricultural Methane: Evidence from Thailand's Rice Sector, 1961–2024

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ABSTRACT

Rice paddies are a major source of methane (CH_4), a greenhouse gas with a 100-year global warming potential (GWP_{100}) of 28. Thailand, one of the world's top rice exporters, cultivates approximately 11 million hectares annually, making accurate CH_4 estimation essential for national climate reporting. This study applies IPCC 2006 Tier 1 at two scales (a national time series 1961–2024 and a 77-province analysis 2022–2024) to quantify rice CH_4 emissions and examine how the *structural design* of price-support programmes determines their emission impact. National emissions averaged **1,420 Gg CH_4 yr^{-1}** (39.8 Mt CO_2e), peaking at 1,884 Gg in 2012 before falling to 1,726 Gg in 2024; in the 2022–2024 provincial analysis applying IPCC Table 5.12 water-regime scaling (SF_w = 0.52 rainfed, 1.0 irrigated) with province-level overrides, the Northeast accounts for 40.1% of provincial CH_4 (482 Gg out of 1,202 Gg point estimate; Monte Carlo 90% CI: 803–1,315 Gg, CV = 14.7%, reflecting uncertainty in SF_w classification and season length). The 2011–2014 Rice Pledging Scheme, an unlimited price guarantee at 40–50% above market, added an estimated 988,000 ha beyond the price effect alone ($p < 0.001$). A counterfactual estimate shows that a per-household area cap applied from 2011 would have avoided approximately **606 Gg CH_4** (17.0 Mt CO_2e) over four years — an upper-bound estimate that illustrates the emission leverage of programme-design choices relative to field-level mitigation technologies. Emission intensity fell 44% (92.5 to 51.4 kg CH_4 t^{-1}) as yields improved, yet absolute emissions rose 84%: a rebound effect in which yield gains raised the profitability of rice cultivation and attracted area expansion. These findings indicate that Thailand's NDC agriculture targets are more sensitive to the *design* of price-support programmes than to yield improvement or field-level mitigation technologies alone.

Keywords: methane emissions; paddy rice cultivation; greenhouse gas inventory; IPCC Tier 1 methodology; Thailand; subnational spatial analysis; agricultural price support; Nationally Determined Contribution; alternate wetting and drying

INTRODUCTION

Thailand is one of the world's great rice economies. Consistently among the top three rice exporters globally (Food and Agriculture Organization of the United Nations, 2024), the country devotes over 11 million hectares to paddy cultivation each year, a landscape that shapes livelihoods, food security, and foreign exchange across Southeast Asia. These same flooded fields also produce a potent greenhouse gas. Kept underwater for months, paddy fields support microbes that generate methane (CH_4), a gas with a GWP_{100} of 28 that has contributed approximately 20% of human-caused radiative forcing since pre-industrial times (Myhre et al., 2013). Agriculture as a whole contributes 10–12% of global greenhouse gas (GHG) emissions, and flooded rice paddies alone account for 25–60 Tg CH_4 yr^{-1}, roughly 5–12% of the global CH_4 budget (Neue, 1993; Yan et al., 2009; Smith et al., 2014).

In flooded fields, microbes break down organic matter without oxygen and release CH_4; the gas escapes to the atmosphere mainly through hollow channels in the rice plant's stem (Neue, 1993). Because emission rates scale directly with flooded area and growing period (Eggleston et al., 2006), any expansion of rice cultivation (more land planted, more seasons per year) directly increases emissions. This study finds that Thailand's paddies emitted on average 1,420 Gg CH_4 yr^{-1} over 1961–2024, and that over the period for which price data are available (2007–2024), farm-gate price policy drives year-to-year area swings, with the 2011–2014 price-support scheme coinciding with the 64-year emission peak.

Despite this striking episode, the quantitative link between rice price support and CH_4 emissions has received limited scientific attention. While broader modelling studies have examined how agricultural subsidies and trade policy reshape land use and GHG emissions across multiple commodities (Frank et al., 2018), few have applied this framework specifically to price-support mechanisms in tropical rice systems. Furthermore, the spatial distribution of Thailand's rice emissions at provincial scale remains poorly documented.

This study pursues three objectives:

1. to quantify the long-term (1961–2024) national CH_4 trend and its statistical characteristics using FAOSTAT data;
2. to map province-scale rice CH_4 emissions and emission intensity across Thailand for 2022–2024 using OAE sub-national data; and
3. to examine the relationship between farm-gate rice price, harvested area, and CH_4 emissions, and discuss implications for Thai rice policy.

DATA AND METHODOLOGY

Study area

Thailand (5°37'–20°27'N, 97°22'–105°38'E) covers 513,120 km^2 and comprises 77 provinces in four macro-regions: North, Northeast, Central, and South. The Northeast Plateau (Khorat Plateau) is the largest rice-producing region, dominated by rainfed single-crop paddy. The Central Plains (Chao Phraya basin) supports double-cropped irrigated paddy with high yields. Northern valleys contain irrigated main-season paddy, while the South cultivates small rice areas amid rubber and oil palm.

Data sources

National activity data (1961–2024). Annual rice harvested area was obtained from FAOSTAT (Food and Agriculture Organization of the United Nations, 2024) (element code 5312), spanning 64 years. FAOSTAT harvested area counts each crop season separately, making it compatible with a per-season emission factor.

Provincial activity data (2022–2024). Sub-national harvested area and production by province were obtained from the Office of Agricultural Economics (OAE) (Office of Agricultural Economics, 2024) for the main season (*khao na pi*) and dry/irrigated season (*khao na prang*) across all 77 provinces. Cross-validation against FAOSTAT national totals for 2022–2024 shows agreement within ±4%, confirming no systematic bias in the OAE provincial data. Data were released as PDF tables; a custom `pdfplumber`-based Python parser was developed to extract and clean records, handling PDF number-splitting artefacts and Thai character encoding. Areas were converted from rai to hectares (1 ha = 6.25 rai).

Rice farm-gate prices (2007–2024). Monthly farm-gate paddy prices (THB per tonne, 15% moisture content) were obtained from OAE (Office of Agricultural Economics, 2024) and converted to annual averages. This price series was merged with FAOSTAT national data for policy analysis.

Spatial boundaries. Province boundaries from GADM 4.1 Level 1 (University of California Davis, 2024) were used for choropleth mapping.

IPCC 2006 Tier 1 emission calculation

CH_4 emissions follow IPCC 2006 Guidelines, Vol. 4, Ch. 5 (Eggleston et al., 2006), Equation 5.1:

$$\mathrm{CH}_{4,i} = \mathrm{EF}_i \times A_i \times 10^{-6} \tag{1}$$

where $\mathrm{CH}_{4,i}$ is annual emission (Gg yr^{-1}), A_i is harvested area (ha), and the seasonal emission factor is:

$$\mathrm{EF}_i = \mathrm{EF}_c \times \mathrm{SF}_w \times \mathrm{SF}_p \times \mathrm{SF}_o \times t_p \tag{2}$$

For the national time-series (1961–2024) this study applies the Tier 1 default $\mathrm{SF}_w = \mathrm{SF}_p = \mathrm{SF}_o = 1.0$ (continuously flooded baseline), yielding EF = 153.4 kg ha^{-1} (main) and 144.3 kg ha^{-1} (dry/irrigated). For the provincial spatial analysis (2022–2024), SF_w is differentiated by region: $\mathrm{SF}_w = 0.52$ for the Northeast and South provinces, and $\mathrm{SF}_w = 1.0$ for continuously flooded irrigated conditions (Central and North). Under IPCC Table 5.12 (Eggleston et al., 2006), $\mathrm{SF}_w = 0.52$ is assigned to the *irrigated, intermittently flooded* (multiple aeration) water regime; the rainfed lowland regular category carries a lower default of $\mathrm{SF}_w = 0.28$. This study applies 0.52 to NE Thailand rainfed provinces following Arunrat and Pumijumnong (2017), who adopted the same value for rainfed fields in Roi-Et province (NE Thailand) on the grounds that seasonal rainfall produces intermittent flooding episodes comparable in drainage frequency to the irrigated multiple-aeration regime. Approximately 74% of Thailand's rice area is rainfed and only 10% of NE rice is irrigated (Climate and Clean Air Coalition, 2021). CO_2-equivalent emissions use GWP_{100} = 28, the AR5 value adopted in Thailand's national inventory reporting. IPCC AR6 subsequently revised GWP_{100} for biogenic CH_4 to 27.9 (Forster et al., 2021), a value already adopted in recent Thai rice GHG studies (Sriphirom and Rossopa, 2023); applying the AR6 value would reduce all CO_2e figures in this study by approximately 0.4%, leaving no qualitative conclusion unchanged.

Table 1. IPCC 2006 Tier 1 parameters used in this study. EF_c from Table 5.11; scaling factors from Tables 5.12–5.14 (Eggleston et al., 2006); $\mathrm{GWP}_{100}(CH_4)$ = 28 (AR5).

Parameter	Value	Source
EF_c	1.30 kg CH_4 ha^{-1} day^{-1}	IPCC Table 5.11
SF_w	1.00 (national); 0.52 NE/South, 1.0 Central/North (provincial)	IPCC Table 5.12
SF_p	1.00 (≤180 days pre-season non-flooded)	IPCC Table 5.13
SF_o	1.00 (no organic amendments)	IPCC Table 5.14
t_p (main season)	118 days	Buddhaboon et al. (2011)
t_p (dry season)	111 days	Buddhaboon et al. (2011)
EF (main)	**153.4 kg ha^{-1} season^{-1}**	Derived
EF (dry)	**144.3 kg ha^{-1} season^{-1}**	Derived
GWP_{100} (CH_4)	28	AR5

Emission intensity and statistical analysis

Emission intensity was calculated as:

$$I_i = \frac{\mathrm{CH}_{4,i} \times 10^6}{P_i} \tag{3}$$

where I_i is emission intensity (kg CH_4 t^{-1} rice) and P_i is rice production (tonnes). Linear regression on annual CH_4 values captures the long-term trend. Emission intensity (kg CH_4 t^{-1} rice) was calculated for 1961–2024 to assess whether higher yields have reduced the climate cost per tonne of rice. Pearson correlation and simple linear regression were used to test the relationship between farm-gate price and harvested area, with statistical significance assessed at $\alpha = 0.05$. As a robustness check, a multivariate OLS model including a binary policy dummy variable ($D_t = 1$ for 2011–2014 Pledging Scheme years, 0 otherwise) was estimated:

$$A_t = \beta_0 + \beta_1 P_t + \beta_2 D_t + \varepsilon_t \tag{4}$$

where A_t is national harvested area and P_t is annual farm-gate price. Residual autocorrelation was assessed via the Durbin–Watson statistic (Durbin and Watson, 1950); Newey–West heteroskedasticity- and autocorrelation-consistent standard errors (Newey and West, 1987) are reported for all OLS models as a background correction to account for potential serial dependence in annual time-series data.

RESULTS AND DISCUSSION

National emission trends and yield paradox (1961–2024)

Figure 1 presents the 64-year time series of harvested area and CH_4 emissions with a fitted linear trend.

Over the full study period, Thailand's mean annual rice CH_4 emission was **1,420 Gg CH_4 yr^{-1}** (39.8 Mt CO_2e). Linear regression reveals a statistically significant upward trend of **+12.0 Gg CH_4 yr^{-1}** ($R^2 = 0.830$, $p = 1.4 \times 10^{-25}$), confirming that emissions have risen substantially over six decades. Emissions grew from 939 Gg in 1961 to a peak of **1,884 Gg in 2012** (an increase of 101%), before declining to 1,726 Gg by 2024. The 2024 value (48.3 Mt CO_2e) represents approximately 13% of Thailand's total national GHG emissions, based on the ˜375 Mt CO_2e total reported in Thailand's Third Biennial Update Report (Thailand Greenhouse Gas Management Organization, 2022).

The long-run growth reflects two overlapping forces. From the 1960s through the 1980s, government investment in irrigation infrastructure and the adoption of high-yielding varieties under Thailand's Green Revolution allowed farmers to cultivate more land and add a second crop per year in irrigated areas. From the 1990s onward, rising global export demand created a sustained incentive to expand production, as Thailand competed with Vietnam and other regional exporters for world market share. Together, these forces drove a steady climb in harvested area and, because CH_4 scales directly with flooded area under Tier 1, in emissions. The 2012 peak stands apart: it was not driven by these long-run structural forces but by a government scheme that made unlimited area expansion immediately profitable (discussed below).

Despite rising absolute emissions, Thailand's emission intensity (CH_4 per tonne of rice produced) has declined sharply over six decades (Figure 2). Intensity fell from **92.5 kg CH_4 t^{-1}** in 1961 to **51.4 kg CH_4 t^{-1}** in 2024, a reduction of 44%, with a highly significant downward trend of −0.730 kg t^{-1} yr^{-1} ($R^2 = 0.892$, $p < 0.001$).

This is a real **food-system efficiency gain**: Thailand now produces each tonne of rice at 44% lower climate cost than in 1961. The improvement reflects nationwide adoption of improved seed varieties

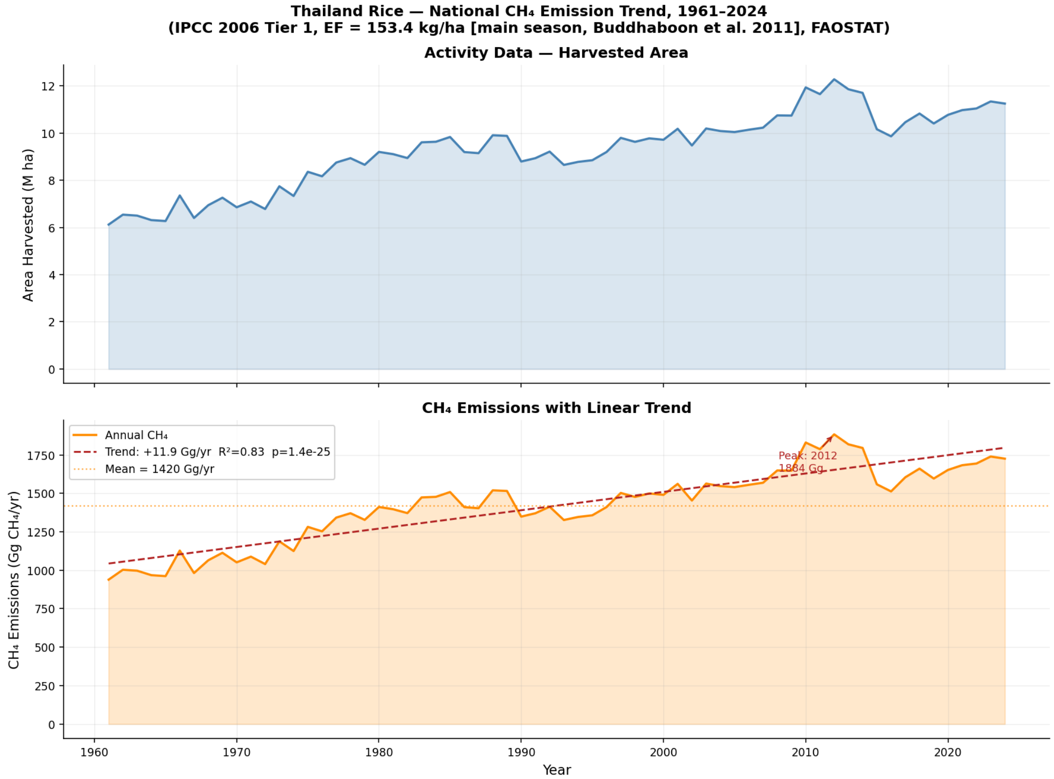

Figure 1. National trends in Thailand rice cultivation, 1961–2024. **Top**: Total harvested area (M ha). **Bottom**: CH_4 emissions (Gg CH_4 yr^{-1}) with linear trend (dashed red) and long-term mean (dotted). Source: Food and Agriculture Organization of the United Nations (2024); IPCC 2006 Tier 1; authors' calculation.

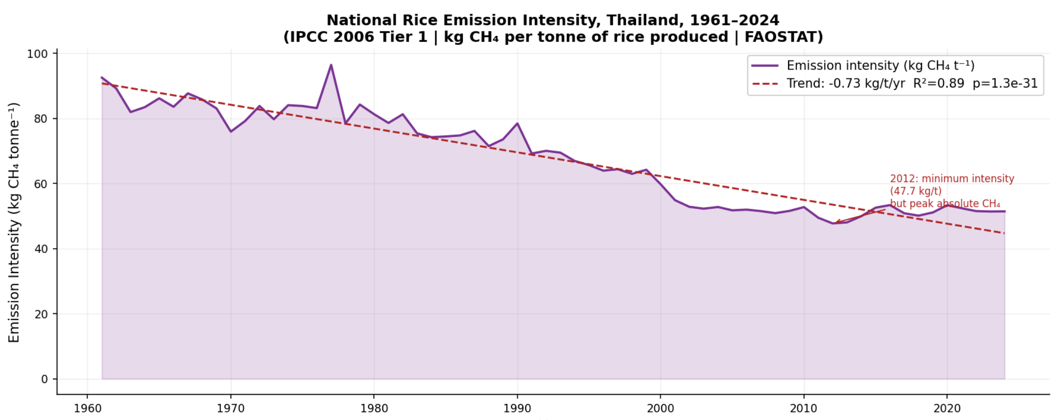

Figure 2. National rice emission intensity (kg CH_4 per tonne of rice produced), Thailand, 1961–2024, with linear trend (dashed red). Intensity has declined 44% over six decades as yield growth outpaced area expansion. Source: Food and Agriculture Organization of the United Nations (2024); IPCC 2006 Tier 1; authors' calculation.

and expanded irrigation, which raised average yields across all regions. Irrigated areas, particularly the Central Plains double-crop system, contributed disproportionately high yields, pulling the national average up and reducing the CH_4 cost per tonne of rice produced. It does **not** mean less methane per hectare. Under Tier 1, intensity is simply $EF_c \times t_p$ divided by yield, so the decline is automatic when yields rise, not a sign of improved field-level water management. Whether actual CH_4 flux per hectare has changed requires Tier 2 field measurements to detect. Finally, total emissions still rose 84% over the same period, because farmers planted more land faster than yields improved. The minimum intensity year was **2012** (47.7 kg t^{-1}): the Pledging Scheme's output incentive drove yields to record levels in Central irrigated areas, inadvertently producing Thailand's most food-efficient harvest on a per-tonne basis, the same year absolute emissions peaked.

This juxtaposition reveals a critical paradox for climate policy: **yield improvement and absolute emission reduction do not automatically co-occur**. Under Tier 1, higher yield reduces emission intensity mechanically (the same CH_4 is spread over more tonnes of rice), but leaves absolute emissions entirely unchanged: only area matters. Beyond the accounting identity, yield gains can actively *increase* absolute emissions through a rebound effect: more profitable production per hectare raises the return to planting, attracting more farmers and pushing marginal land into cultivation. Thailand's 64-year record illustrates this precisely: the Green Revolution raised both yields *and* planted area simultaneously, so absolute emissions climbed even as intensity fell. The implication for policy is direct: programmes that raise on-farm yields without constraining area (through caps, zoning, or purchase limits) may inadvertently drive emission increases. Yield improvement reduces absolute CH_4 only if it enables the *same total production on less land*, a land-sparing outcome that requires deliberate area management, not just better agronomy.

Spatial distribution of the climate cost

Figure 3 maps mean annual CH_4 emissions per province for 2022–2024. The Northeast clearly dominates the spatial pattern, consistent with its large paddy area on the Khorat Plateau.

Table 2. Top 10 rice-CH_4-emitting provinces, Thailand, mean annual 2022–2024 (IPCC 2006 Tier 1; SF_w differentiated: 0.52 rainfed NE/South, 1.0 irrigated Central/North).

Rk	Province	Region	Area (kha)	CH_4 (Gg yr^{-1})	Intensity (kg t^{-1})
1	Nakhon Sawan	Central	478	72.4	40.5
2	Phichit	North	368	55.7	39.4
3	Ubon Ratchathani	Northeast	662	52.7	34.7
4	Phitsanulok	North	331	49.9	40.2
5	Suphan Buri	Central	331	49.6	33.6
6	Nakhon Ratchasima	Northeast	564	44.9	32.8
7	Surin	Northeast	491	39.1	33.2
8	Roi Et	Northeast	490	38.9	34.7
9	Sukhothai	North	258	38.9	42.2
10	Chiang Rai	North	253	38.3	42.6

kha = thousand ha; intensity = kg CH_4 per tonne rice produced.

Table 2 lists the ten highest-emitting provinces under the SF_w-differentiated calculation. **Nakhon Sawan** (Central, 72.4 Gg yr^{-1}) now leads, reflecting its large double-cropped irrigated area (SF_w = 1.0) and high yields. **Phichit** (North, 55.7 Gg) ranks second for the same reason. **Ubon Ratchathani** (Northeast, 52.7 Gg) falls to third: despite having the largest cultivated area in the country (662 kha), its rainfed regime (SF_w = 0.52) halves the per-hectare emission factor. Nakhon Sawan's position as the top emitter is

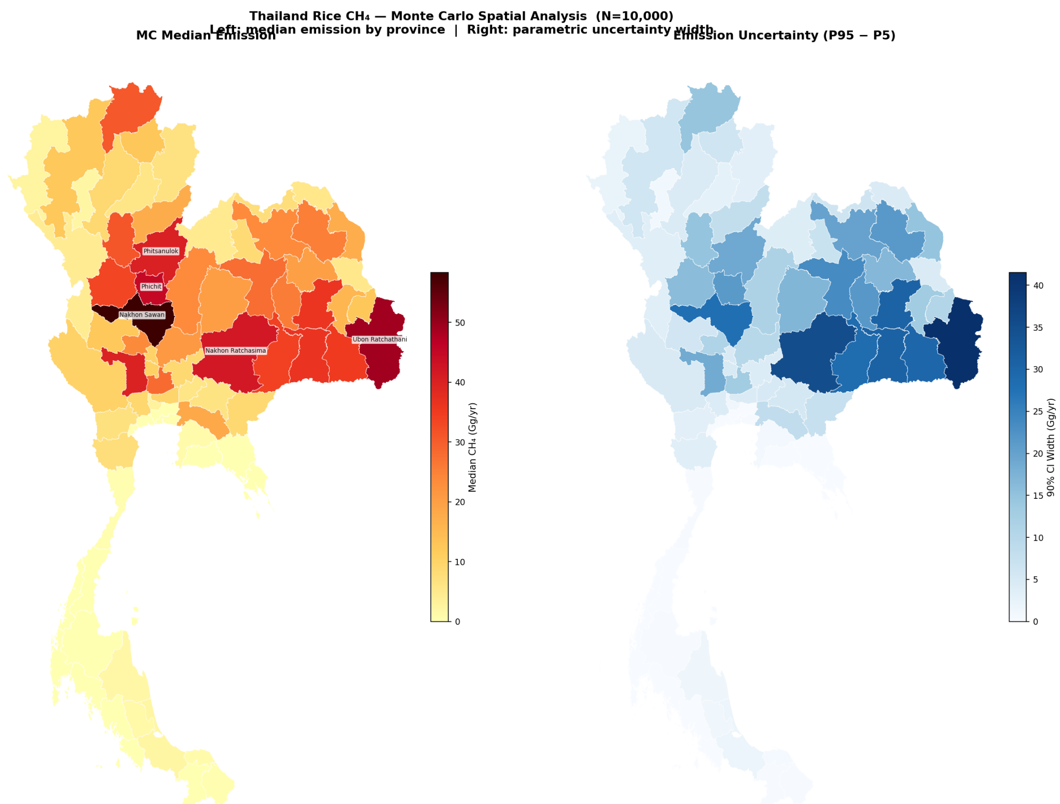

Figure 3. Monte Carlo spatial analysis of mean annual rice CH_4 emissions by province, Thailand, 2022–2024 (N = 10,000). **Left:** MC median emission (Gg yr^{-1}); pale yellow = low, near-black = high; top-5 emitting provinces labelled. **Right:** 90% CI width (P95 − P5), reflecting parametric uncertainty from SF_w and t_p; deep blue = highest absolute uncertainty. Large irrigated provinces (Central core, North) show the widest CI because SF_w variance scales with emission magnitude. Source: Office of Agricultural Economics (2024); University of California Davis (2024); authors' calculation.

robust: its 90% CI spans only ranks 1–3 across all $N = 10{,}000$ draws, because its irrigated area is large enough that no plausible SF_w draw allows another province to displace it consistently.

After SF_w differentiation and province-level overrides, the Northeast accounts for **40.1%** (482.0 Gg), while the Central Plains (**31.3%**, 376.1 Gg) and North (**27.4%**, 329.9 Gg) together account for 58.7%. The South contributes 1.2% (14.0 Gg). Regional SF_w defaults follow IPCC Table 5.12 (0.52 rainfed NE/South; 1.0 irrigated Central/North), with nine peripheral provinces overridden to $SF_w = 0.52$: the eastern-Central provinces of Sa Kaeo, Chanthaburi, Trat, Prachin Buri, Rayong, Chon Buri, and Nakhon Nayok (outside the Chao Phraya irrigation network), and the remote northern provinces of Mae Hong Son and Tak (highland mountain rice). The main wet season dominates emissions in all regions; the dry season contributes a larger-than-expected share in the Central Plains, where year-round irrigation supports a second rice crop (seasonal breakdown in Appendix Figure 8 and Table 4).

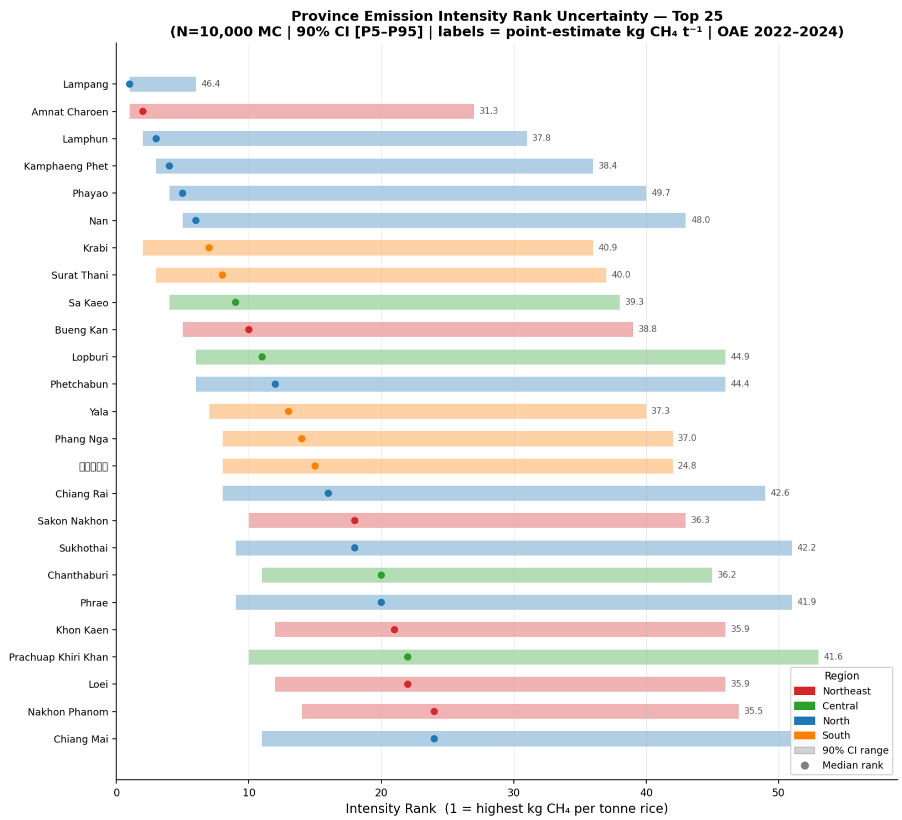

Figure 4. Top 25 provinces ranked by emission intensity (kg CH_4 t^{-1} rice produced), with Monte Carlo rank uncertainty ($N = 10{,}000$). Bars show 90% CI [P5–P95] of intensity rank; dots show median rank (rank 1 = highest intensity). Labels give point-estimate intensity. Wide CIs reflect sensitivity to the rainfed/irrigated SF_w ratio: the intensity ranking is less stable than the absolute-emission ranking because the rainfed–irrigated yield differential interacts with the SF_w draw to reshuffle province order. Source: Office of Agricultural Economics (2024); authors' calculation.

Figure 4 reveals a striking reversal from the total-emission picture once water-regime differences are properly accounted for. The Northeast dominates national CH_4 *by volume* (40.1%, 482 Gg), yet its mean emission intensity (**34.0 kg t^{-1}**) is nearly identical to the Central Plains (**34.1 kg t^{-1}**). Both trail the **North** (**41.0 kg t^{-1}**), which carries the highest regional intensity.

The mechanism is clear in the top of the intensity chart: the leading provinces are all single-crop irrigated North provinces—**Phayao** (49.7 kg t^{-1}), **Nan** (48.0), **Lampang** (46.4)—that receive $SF_w = 1.0$

(continuously flooded) but produce only one harvest per year. Their per-hectare CH_4 flux is high ($SF_w = 1.0 \times EF_c \times t_p$), yet their single-season yield base is modest; the ratio is therefore higher than in Central double-crop provinces (e.g. Nakhon Sawan: 40.5 kg t^{-1}), where two harvests per year spread the same per-hectare cost across twice the production. The convergence of NE and Central intensity (≈34 kg t^{-1}) is a direct consequence of correcting SF_w: rainfed NE paddies emit half as much CH_4 per hectare as irrigated Central paddies, and this reduction exactly offsets the NE yield disadvantage at the regional average.

In short: **volume and efficiency tell different stories, and both matter for policy.** The Northeast is the *quantity* target (largest absolute emissions); the irrigated North is the *efficiency* target (highest CH_4 per tonne of rice produced). These spatial patterns, however, are not fixed: price policy can drive sharp year-to-year swings in cultivated area and thus total emissions.

Econometrics: price policy as an emission driver

Geography explains *where* Thailand's rice emissions come from; price policy explains *when* they spike. Figure 5 overlays farm-gate paddy price, yield, and estimated CH_4 emissions from 2007 to 2024, annotated with key policy events; the co-movement between price and CH_4 is immediate and striking.

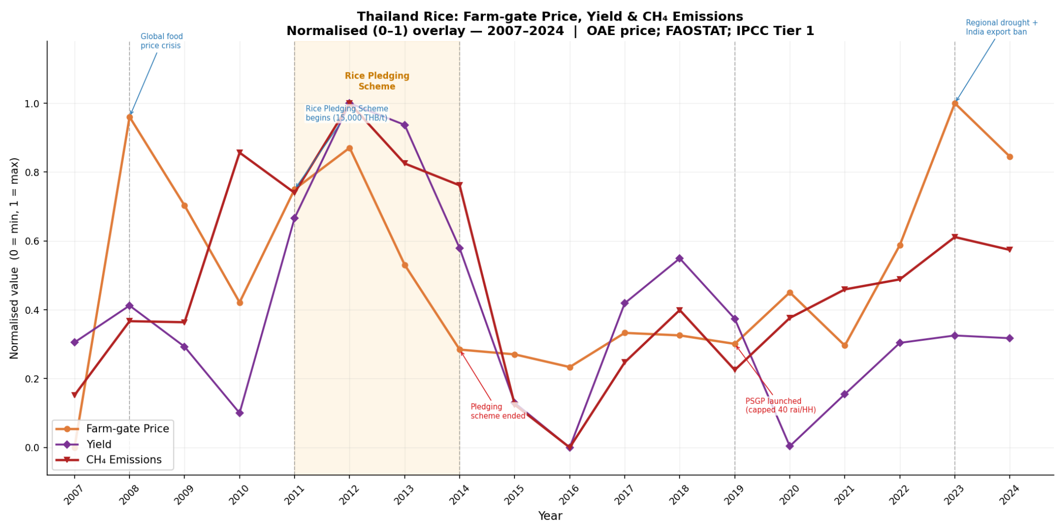

Figure 5. Thailand rice farm-gate price (THB $tonne^{-1}$), yield (hg ha^{-1}), and estimated CH_4 emissions (Gg yr^{-1}), normalised (0–1), 2007–2024. Vertical dashed lines mark major policy events. Sources: Office of Agricultural Economics (2024) (price); Food and Agriculture Organization of the United Nations (2024); authors' calculation.

The price time series shows three distinct periods. During **2008**, the global food price crisis briefly doubled farm-gate prices (6,600 to 10,500 THB $tonne^{-1}$), triggering a modest area expansion. The most pronounced episode is the **Rice Pledging Scheme** (*khrongkan rap chamnam khao*; 2011–2014), introduced as an electoral commitment by the Pheu Thai government. The scheme obliged the government to purchase *all* paddy offered by farmers at 15,000 THB $tonne^{-1}$ for white rice and 20,000 THB $tonne^{-1}$ for jasmine, approximately 40–50% above world market prices, with **no quantity cap per household** (Poapongsakorn and Pantakua, 2014). This design created an unlimited expansion incentive: every additional tonne planted was guaranteed to be profitable regardless of market conditions. Farmers responded rationally by converting non-rice land to paddy, expanding double-cropping in irrigated areas, and pushing marginal fields into production. National harvested area reached record levels and CH_4 reached its 64-year peak of **1,884 Gg in 2012**.

The scheme ultimately became fiscally unsustainable. The government purchased approximately 52% of total national paddy production but could not resell stockpiles on world markets: Thai offer prices were too high relative to competing exporters (USDA Foreign Agricultural Service, 2017). Warehoused rice deteriorated, widespread corruption in storage contracts emerged, and the total programme cost reached **984 billion THB** (approximately USD 32 billion), equivalent to 41% of the 2013 national budget (Poapongsakorn and Pantakua, 2014; USDA Foreign Agricultural Service, 2017). Following the May 2014 military coup, the incoming government cancelled the scheme; farm-gate prices fell sharply and harvested area contracted by over 5% in 2014/15, pulling CH_4 downward proportionally. This sequence—policy introduced, emissions spike, policy cancelled, emissions fall—is consistent with a causal interpretation, though formal proof would require a counterfactual comparison region or a more complete model.

Since 2015, the government has operated a lighter-touch **Price Subsidy Guarantee Programme** (PSGP), which pays farmers the difference between a reference price and the market price, capped at 40 rai (6.4 ha) per household. This redesign shows a key principle: *how* a price-support programme is structured matters as much as the price level itself. The Pledging Scheme's unlimited guarantee made every extra hectare profitable no matter what, a direct incentive to expand. The PSGP's per-household cap removes that incentive beyond 6.4 ha, limiting the land-expansion response. Since 2019, price and area trajectories have been more stable, showing that climate-friendly policy design does not require abandoning farmer income support.

The counterfactual: emission costs of uncapped guarantees

Counterfactual: the emission cost of programme design. The regression dummy coefficient ($\hat{\beta}_2$ = 988 kha) isolates the structural expansion effect of the unlimited guarantee, the excess area beyond what the contemporaneous farm-gate price alone would predict. This provides a direct basis for a counterfactual: *had the Pledging Scheme included the same 40-rai (6.4 ha) per-household cap as the current PSGP, this structural expansion would have been constrained.* Applying the main-season emission factor, the estimated **avoided CH_4 is 151.6 Gg yr^{-1}**, equivalent to 10.7% of the long-run national mean, for each year the cap would have been in effect. Summed across the scheme's four-year duration (2011–2014), the cap would have prevented approximately **606 Gg CH_4** (17.0 Mt CO_2e). These findings indicate that the structural design of price-support programmes operates at a scale of emission impact that field-level mitigation technologies alone are unlikely to match. The SF_w and season-length uncertainty propagated through the MC does not affect this figure, since the counterfactual is computed from the national area regression using the fixed Tier 1 EF. The counterfactual should be read as an illustrative upper-bound estimate, not a reliable point estimate, for three reasons. First, the dummy coefficient ($\hat{\beta}_2$ = 988 kha, n = 18) is estimated from a short time series with omitted variables (rainfall, export demand, input costs) that are correlated with both price and the scheme dummy; the true structural expansion effect could be considerably smaller or larger. Second, the cap would bind only for households already operating above 6.4 ha; smallholders below that threshold would be unaffected and their price-driven expansion would continue. Third, capped subsidy schemes in Thailand have documented instances of land-title subdivision, where households divide holdings among family members to remain under the cap threshold, a behavioural response that would partially erode the cap's effectiveness in practice (Poapongsakorn and Pantakua, 2014). The 2023 spike in price (10,700 THB $tonne^{-1}$) reflects regional drought reducing supply combined with elevated global export demand following the India rice export ban.

Statistical analysis confirms a **significant positive relationship** between farm-gate price and harvested area (r = 0.505, p = 0.033, n = 18). A regression slope of 0.045 Gg CH_4 per THB $tonne^{-1}$ (derived from the area–price slope of 291 ha per THB $tonne^{-1}$) implies that a 1,000 THB $tonne^{-1}$ sustained price increase is associated with approximately 45 Gg additional CH_4 yr^{-1}, equivalent to about 2.6% of the 2024 national total.

Table 3. Regression of national harvested area on price and Pledging Scheme dummy, Thailand, 2007–2024 (n = 18). Dependent variable: national harvested area (ha). D_t = 1 for 2011–2014, 0 otherwise. HAC standard errors use Newey–West estimator (maxlags = 1) (Newey and West, 1987).

	Model 1 *Price only*	**Model 2** *Price + dummy*	**Model 2 (HAC)** *NW SE*
Intercept	8,490,000 ha	8,869,000 ha	8,869,000 ha
Price ($\hat{\beta}_1$, ha/THB t^{-1})	291	222	222
p-value	0.033	0.029	**0.005**
Pledging dummy ($\hat{\beta}_2$, kha)	—	+988	+988
p-value	—	0.001	<**0.001**
R^2	0.255	0.631	—
ΔR^2	—	+0.376	—
Durbin–Watson	1.162	2.141	—

CH_4 equivalent of $\hat{\beta}_2$: 151.6 Gg yr^{-1} (EF = 153.4 kg ha^{-1}, Buddhaboon et al. 2011).
DW ≈ 2 for Model 2 indicates autocorrelation resolved once dummy is included.

To test whether the Pledging Scheme expanded area beyond the price effect alone, a policy dummy (D_t = 1 for 2011–2014) was added to the regression (Equation 4; Table 3). Adding the dummy raises R^2 from 0.255 to 0.631. The dummy coefficient ($\hat{\beta}_2$ = 988 kha) shows that the Pledging Scheme years are linked to roughly **988,000 additional hectares** beyond what price alone explains, equivalent to **151.6 Gg CH_4 yr^{-1}** (4,245 Gg CO_2e). The price slope also drops from 291 to 222 ha per THB tonne^{-1} once the dummy is included, showing the simpler model was blending the scheme's direct area effect with the general price response. Both effects remain highly significant under Newey–West HAC standard errors (Newey and West, 1987) ($p_{\text{price}} = 0.005$; $p_{\text{pledging}} < 0.001$). The dummy coefficient should be read as capturing the scheme's *combined* effect (both the price premium and the structural guarantee of unlimited government purchasing) rather than a clean separation of the two, given that D_t and P_t are correlated during the 2011–2014 pledge years.

Forward-looking mitigation: targeting AWD in irrigated systems

These findings have direct policy implications. First, **price-support programmes must account for their emission co-effects**: purchasing guarantees that raise farm-gate prices above equilibrium will expand the cultivated area and increase CH_4 proportionally. Emission impact assessments should be a standard component of rice subsidy design under Thailand's NDC framework. Second, **targeted adoption of Alternate Wetting and Drying (AWD)**, which involves intermittent drainage during the growing season, could cut per-hectare CH_4 by 30–50% without yield loss (Wassmann et al., 2000; Maneepitak et al., 2019; Sriphirom and Rossopa, 2023). A suitability assessment across six Central Thai provinces (Ang Thong, Ayutthaya, Chai Nat, Pathum Thani, Sing Buri, and Suphan Buri) estimated annual CH_4 reductions of 32% under AWD adoption—24% in the major wet season and 41% in the dry season (Prangbang et al., 2020). AWD is applicable only to the dry-season irrigated area, where water control infrastructure enables scheduled drainage; only 25% of Thailand's rice area is irrigated (Climate and Clean Air Coalition, 2021), limiting AWD's national reach to the Central and North provinces. Province-level analysis of dry-season harvested area (Figure 6) identifies **Suphan Buri** (Central), **Phitsanulok** (North), **Phra Nakhon Si Ayutthaya** (Central), and **Nakhon Sawan** (Central) as the highest-priority provinces, together accounting for 35% of national AWD mitigation potential. Applying a conservative 30% EF

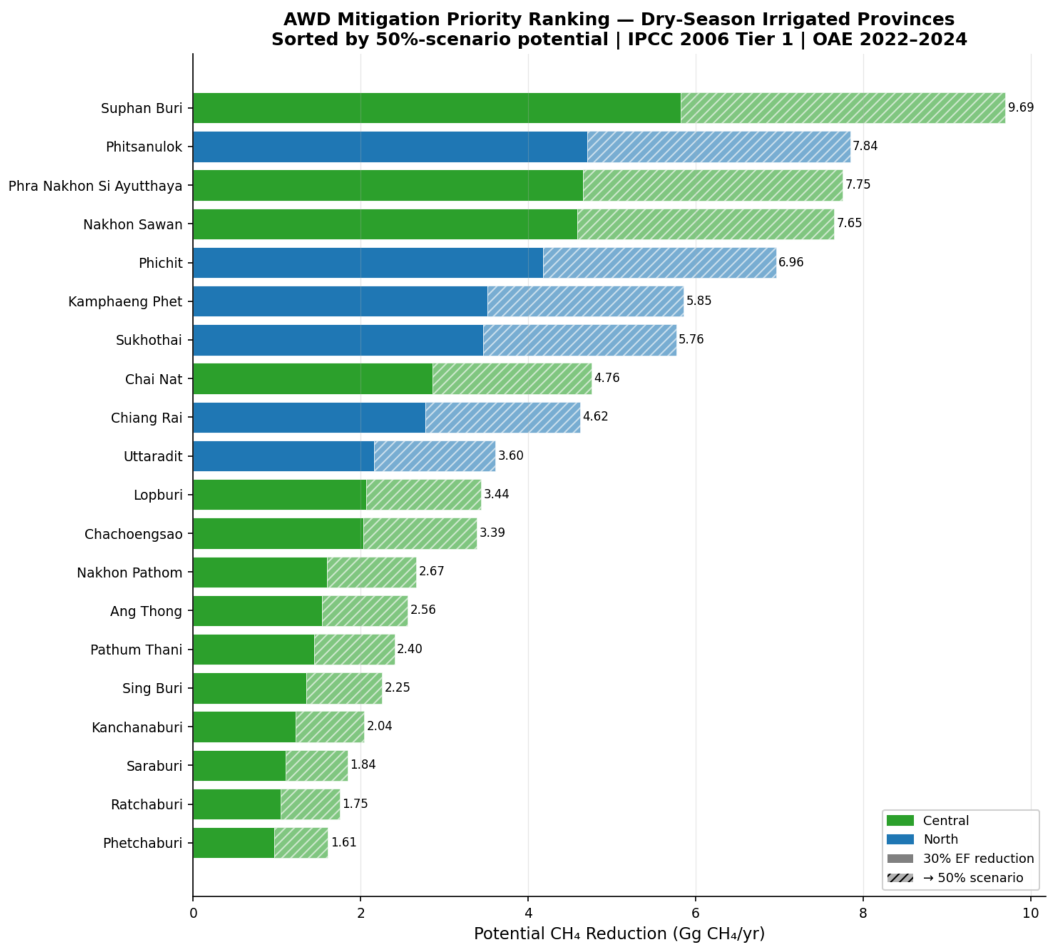

Figure 6. AWD mitigation priority ranking by province. Bars show potential CH_4 reduction under 30% (solid) and 50% (hatched extension) EF reduction scenarios applied to dry-season irrigated area only. Bar colour indicates macro-region. Source: Office of Agricultural Economics (2024); authors' calculation.

reduction to irrigated dry-season provinces yields a national reduction of **57.3 Gg yr^{-1}** (3.3% of the 2024 total); a 50% reduction saves **95.5 Gg yr^{-1}** (5.5%). Extending AWD incentives through carbon credit mechanisms tied to Thailand's NDC represents a priority opportunity (Royal Thai Government, 2022). Critically, Thailand's evolving price-support architecture offers a direct implementation lever: embedding AWD compliance as a prerequisite for PSGP subsidy payments would align income transfers with climate outcomes, delivering CH_4 reductions precisely in the irrigated areas where water control is feasible and programme monitoring is most practicable.

Four concrete policy steps follow from this analysis. First, AWD adoption should be **prioritised** in Suphan Buri, Phitsanulok, Phra Nakhon Si Ayutthaya, and Nakhon Sawan, the four provinces that together account for the largest share of national AWD mitigation potential and already operate the irrigation infrastructure required for scheduled drainage. Second, AWD compliance should be made a **mandatory condition** for PSGP subsidy receipt rather than a voluntary option, using OAE's existing plot-level monitoring to verify adoption without creating a parallel bureaucracy. Third, verified AWD emission reductions should be linked to Thailand's **domestic carbon market** or international climate finance instruments under the NDC framework (Royal Thai Government, 2022), giving irrigated-area farmers a direct monetary incentive that complements the income guarantee. Together, these steps would use the same PSGP administrative architecture that already caps area expansion to additionally mandate cleaner water management, turning a single programme into a dual lever for income support and climate compliance. Fourth, the **North** carries the highest emission intensity (41.0 kg t^{-1}) because its provinces are single-crop continuously flooded but produce only one harvest per year. **AWD adoption in North irrigated provinces** therefore addresses both the efficiency and absolute emission dimensions simultaneously—cutting the per-hectare CH_4 flux while maintaining yield. For the Northeast, where SF_w = 0.52 already reflects natural drying cycles, **drought-tolerant variety programmes** that raise yields would reduce intensity without requiring changes to water management infrastructure.

Limitations and uncertainties

The biggest source of uncertainty in this study is probably the water management assumption. The provincial analysis assigns SF_w = 0.52 to the Northeast and South rainfed provinces and SF_w = 1.0 for continuously flooded irrigated conditions (Central and North)—an improvement over the uniform Tier 1 baseline but still an approximation. Under IPCC Table 5.12, 0.52 is the *irrigated* intermittently flooded factor; the rainfed regular default is 0.28. Within the Northeast, flooding regimes vary considerably: deeper-flood zones (e.g. the Mun River basin) may sustain near-continuous inundation, while drier upland rainfed areas are closer to the rainfed regular regime (SF_w = 0.28), suggesting the true provincial average lies somewhere in the 0.28–0.52 range. Using a single regional value of 0.52 therefore remains a simplification; Northeast estimates presented here are best read as regional averages rather than precise provincial values. For Central Thailand specifically, field experiments in Ayutthaya found that AWD irrigation reduces CH_4 by 34–39% versus continuous flooding (Maneepitak et al., 2019), implying an effective SF_w in the range 0.61–0.66 under water-saving management and suggesting the IPCC default of 1.0 may overestimate actual Central emissions by 30–40%. This uncertainty is captured in the Monte Carlo analysis via a wider SF_w irrigated range of [0.61, 1.00]. The national time-series estimates (1961–2024) use the standard Tier 1 value SF_w = 1.0, consistent with national inventory reporting practice, and should be interpreted as a conservative upper bound.

A more fundamental constraint is the Tier 1 formula itself. Because $CH_4 = EF \times \text{Area}$ and neither the emission factor nor the scaling factors respond to price signals, any policy-driven area expansion produces an exactly proportional emission increase by construction. The observed correlation between price support and emissions is therefore partly a methodological artifact of the linear inventory formula, not direct evidence of a real-world biophysical change. A Tier 2 approach — in which field-measured EFs vary with actual water management, organic inputs, and variety choice — could produce a different relationship if higher-price years systematically alter management practices (e.g. greater fertiliser use or more intensive flooding to boost yields). This study's policy conclusions are directionally robust under Tier 1 because the causal mechanism (price → area expansion → more flooded hectares) is real and empirically supported; but the quantitative emission figures should be understood as inventory constructs rather than independent physical measurements.

The uniform cultivation period (118 days main season, 111 days dry season, Buddhaboon et al. 2011) is a second simplification. Real season lengths vary by province and variety (Buddhaboon et al. 2011 report σ = 5 days for main season and σ = 6 days for dry season across varieties and planting dates), and province-level data to differentiate them remain limited.

To formally propagate the key parametric uncertainties, Figure 7 presents a Monte Carlo analysis (N = 10,000) varying four parameters: SF_w rainfed (uniform, 0.27–0.71), SF_w irrigated (uniform, 0.61–1.00), and both season lengths (normal, σ = 5 days main season and σ = 6 days dry season, derived from Buddhaboon et al. 2011 Table 1 across varieties and planting dates). EF_c is held fixed at the IPCC 2006 Tier 1 default of 1.30 kg ha^{-1} day^{-1} (Eggleston et al., 2006); the analysis is designed to quantify uncertainty from the regional SF_w grouping assumption (two classes for 77 provinces), not from the emission factor itself. The lower bound of 0.61 for irrigated SF_w is derived from field experiments in Ayutthaya, Central Thailand, where AWD reduced CH_4 by 34–39% versus continuous flooding (Maneepitak et al., 2019), while the upper bound of 1.00 retains the IPCC default for continuously flooded conditions.

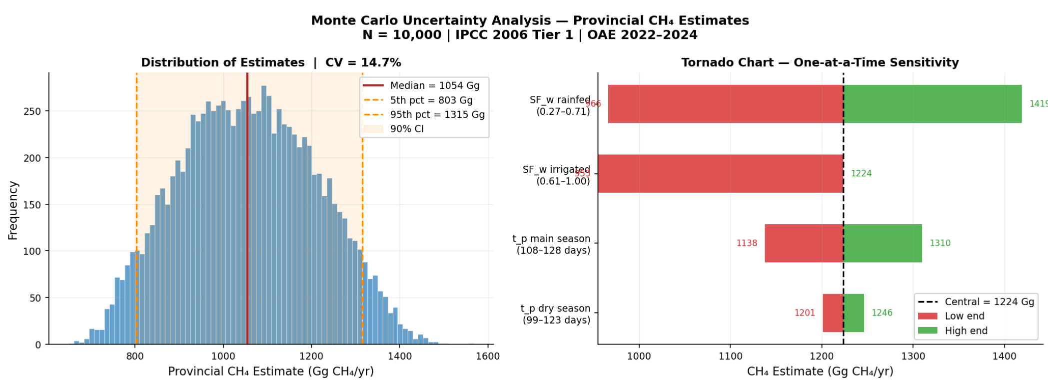

Figure 7. Monte Carlo uncertainty analysis (N = 10,000) for provincial CH_4 estimates. *Left:* distribution of simulated totals; median = 1,054 Gg, 90% CI [803, 1,315] Gg, CV = 14.7%. *Right:* tornado chart — one-at-a-time sensitivity showing SF_w rainfed as the dominant source of uncertainty, confirming that the regional water-regime classification drives model output more than season length. EF_c is fixed at the IPCC Tier 1 default (1.30). Irrigated SF_w lower bound [0.61] from Maneepitak et al. (2019); upper bound 1.00 is the IPCC default. Source: authors' calculation; IPCC 2006 (Eggleston et al., 2006).

With EF_c fixed, the distribution is near-symmetric with a coefficient of variation of **14.7%**, substantially tighter than the ~50% uncertainty reported in global rice CH_4 inventories that also vary EF_c (Yan et al., 2009). This 14.7% CV reflects *classification uncertainty only* — the uncertainty in SF_w regional grouping and season length — and should not be interpreted as total inventory uncertainty; allowing EF_c to vary over its published range would raise the CV to approximately 45–50% (Yan et al., 2009). The tornado chart confirms that **SF_w rainfed dominates** (966–1,419 Gg), because 43 of 77 provinces draw from the rainfed uniform distribution and the NE accounts for 40% of total area. Main-season length is second

(1,136–1,308 Gg), irrigated SF_w is third (955–1,224 Gg), and dry-season length contributes negligibly (1,199–1,244 Gg). This hierarchy directly validates the analytical focus of this paper: the largest reducible uncertainty is the two-class SF_w grouping; replacing it with province-specific flooding-regime data (Tier 2) would narrow the CI more than any other methodological improvement. The policy conclusions are unaffected by this uncertainty: the trend direction, the Pledging Scheme significance ($p < 0.001$), and regional emission shares are derived from area data and regression slopes and do not depend on SF_w. Province-level emission *ranks* carry additional uncertainty beyond the aggregate total: because EF_c and t_p scale all provinces identically, only the rainfed/irrigated SF_w ratio can reshuffle the rank order. A province-level Monte Carlo (same N = 10,000 draws, applied as a $76 \times N$ emission matrix) shows that **Nakhon Sawan**'s top rank is stable (90% CI: ranks 1–3), while mid-table Northeast and North provinces may shift by 10–20 positions.

Field-measured, province-specific emission factors would address the Tier 1 uncertainties on the methods side (Wassmann et al., 2000; Yan et al., 2009). A natural extension is a province-level Tier 2 study applying field-measured SF_w values rather than the regional categorical defaults used here. Such a study would replace the two-class (0.52/1.0) approximation with province-specific flooding-regime data, further refining the spatial pattern. Under Tier 2, provinces at the wetter end of the NE distribution (deeper-flood areas) might see SF_w approach 0.8–1.0, while drier upland rainfed areas could be as low as 0.28. Crucially, the North and Central Plains—identified here as the highest-intensity irrigated regions (SF_w = 1.0, North intensity 41.0 kg t^{-1})—would retain elevated SF_w values under Tier 2, validating the AWD priority ranking derived in this study. Finally, while the policy section proposes embedding AWD compliance as a condition for PSGP subsidy receipt, implementing this credibly requires monitoring infrastructure beyond what currently exists: scheduled drainage must be documented through water-level sensors, soil moisture records, or sufficiently high-resolution remote sensing, none of which OAE's current plot-area surveys provide. MRV build-out costs should therefore be evaluated against mitigation benefits before mandatory conditionality is adopted.

CONCLUSIONS

Thai rice paddies are a significant and growing source of agricultural methane, but one whose trajectory is shaped more by the structural design of price-support programmes than by either agronomic improvement or field-level mitigation technology. The central finding of this study is that a single design choice (the absence of a per-household area cap in the 2011–2014 Rice Pledging Scheme) is estimated to have generated 606 Gg CH_4 (17.0 Mt CO_2e) of avoidable emissions over four years — an upper-bound estimate illustrating the emission leverage of programme-design decisions. The practical implication is that Thailand's path to meeting its NDC agriculture targets runs primarily through the design of its next rice price-support programme, not through yield improvement or AWD alone. The key findings of this study are:

1. National emissions increased significantly at +12.0 Gg yr^{-1} ($R^2 = 0.830$, $p < 0.001$) over 1961–2024, peaking at 1,884 Gg in 2012 and reaching 1,726 Gg (48.3 Mt CO_2e) in 2024.
2. Applying IPCC Table 5.12 SF_w with province-level overrides for nine peripheral rainfed provinces, the Northeast accounts for 40.1% (482.0 Gg yr^{-1}), Central 31.3% (376.1 Gg), and North 27.4% (329.9 Gg); **Nakhon Sawan** (Central, 72.4 Gg) is the highest-emitting single province. A Monte Carlo uncertainty analysis (N = 10,000) yields a 90% CI of [803, 1,315] Gg for the provincial total, with CV = 14.7%; SF_w rainfed is the dominant uncertainty driver, confirming that province-level water-regime classification is the binding methodological constraint.

3. National emission intensity declined from 92.5 to 51.4 kg CH_4 t^{-1} (−44%; $R^2 = 0.892$, $p < 0.001$), reflecting a genuine food-system efficiency gain as yield growth outpaced area expansion. This is *relative* decoupling only: absolute emissions rose 84% over the same period, and per-hectare flux is unchanged under Tier 1's constant EF.
4. After SF_w correction with province-level overrides, the **North** carries the highest mean intensity (41.0 kg CH_4 t^{-1}), driven by single-crop irrigated provinces; the Central Plains and Northeast converge at ≈34 kg t^{-1} once eastern-Central rainfed provinces are correctly reclassified ($SF_w = 0.52$). AWD targets irrigated provinces where $SF_w = 1.0$ elevates per-hectare flux.
5. Farm-gate price is a significant positive predictor of harvested area ($r = 0.505$, $p = 0.033$, $n = 18$); the 2011–2014 Rice Pledging Scheme added ~988,000 ha beyond the price effect alone, consistent with a policy-induced area expansion driving the 2012 emission peak.
6. **Programme design determines emission risk more than price level.** A counterfactual cap-equivalent calculation estimates that a 40-rai per-household limit applied from 2011 would have avoided **606 Gg CH_4** (17.0 Mt CO_2e) over the scheme's four-year duration (upper-bound estimate). Future price-support programmes should treat area cap design as a primary climate-policy variable, not an afterthought.
7. **Yield improvement does not automatically reduce absolute emissions.** Emission intensity fell 44% over 1961–2024 as yields improved, yet absolute emissions rose 84% because yield gains raised the profitability of rice farming and attracted area expansion (rebound effect). The 2012 emission peak coinciding with Thailand's lowest-ever emission intensity (47.7 kg t^{-1}) illustrates this directly. Absolute CH_4 reduction requires either area contraction or water management change; yield improvement alone is insufficient and may be counterproductive when paired with uncapped price guarantees.
8. AWD adoption across dry-season irrigated provinces reduces national CH_4 by 3–6% under 30–50% EF reduction scenarios (57.3 Gg and 95.5 Gg yr^{-1} respectively), with irrigated Central and North provinces (led by Suphan Buri [Central] and Phitsanulok [North]) accounting for the largest share of mitigation potential. Embedding AWD compliance as a condition for PSGP subsidy receipt represents the most actionable mitigation pathway under Thailand's NDC.

ACKNOWLEDGEMENTS

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AI USE DISCLOSURE

Claude (Anthropic), an AI language model, was used during the preparation of this manuscript to assist with prose editing, structural feedback, and LaTeX formatting. All data collection, emission calculations, statistical analysis, and intellectual conclusions are the author's own. The author takes full responsibility for the accuracy and integrity of the content presented.

APPENDIX

These appendices document the full provincial dataset and highlight a key divergence between production volume and emission intensity. The Northeast dominates rice output (~14.2 Mt yr^{-1}, 43% of national production) but its rainfed water regime (SF_w = 0.52) keeps emission intensity at ≈34 kg CH_4 t^{-1}—essentially equal to the irrigated Central Plains. The North achieves the highest intensity (41.0 kg t^{-1}) despite ranking third in absolute output, because single-crop irrigated systems combine full flooding (SF_w = 1.0) with moderate yields (~3.45 t ha^{-1}). Small North provinces such as Phayao (49.7 kg t^{-1}) and Nan (48.0 kg t^{-1}) rank among the most emission-intensive but contribute little to total volume.

Table 4. Regional rice CH_4 emission summary, mean annual 2022–2024 (IPCC 2006 Tier 1, SF_w-differentiated; † marks provinces with rainfed override applied within administrative region).

Region	**Provs**	**Area** (kha)	**Prod.** (Mt)	**Yield** (t ha^{-1})	**CH_4** (Gg)	**CO_2e** (Gg)	**Share** (%)	**Int.** (kg t^{-1})
Northeast	20	6,060	14.18	2.35	482.0	13,497	40.1	34.0
Central	28	2,648	10.45	3.87	376.1	10,531	31.3	34.1
North	15	2,225	8.01	3.45	329.9	9,239	27.4	41.0
South	14	178	0.54	2.61	14.0	393	1.2	31.4
Total	**77**	**11,110**	**33.16**	—	**1,202.1**	**33,660**	**100.0**	—

SF_w: NE/South = 0.52; Central/North = 1.0 (IPCC Table 5.12); 9 override provinces = 0.52.

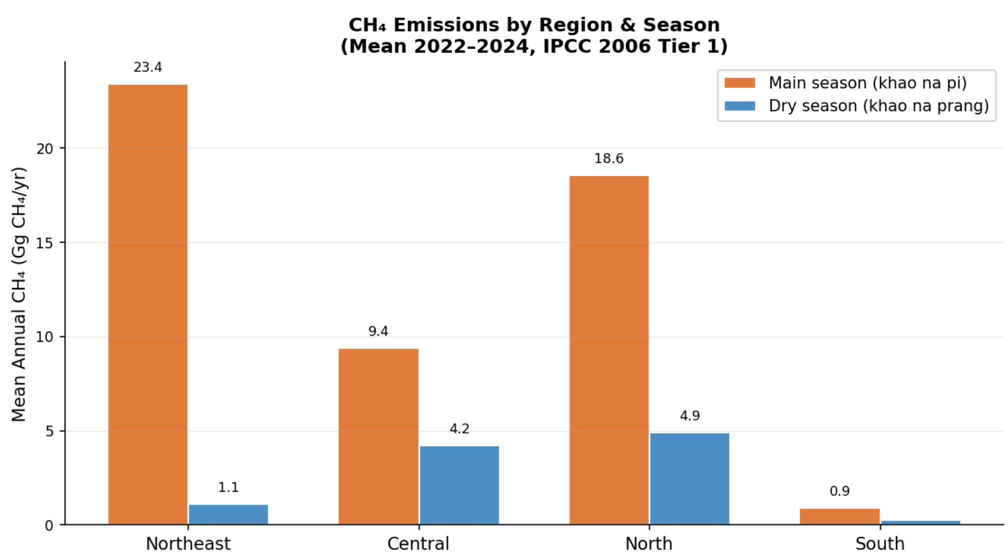

Figure 8. Mean annual rice CH_4 emissions by region and season, Thailand, 2022–2024. Orange: main season (*khao na pi*); blue: dry season (*khao na prang*). Source: Office of Agricultural Economics (2024); authors' calculation.

Table 5. Top 20 provinces by rice CH_4 emission intensity (kg CH_4 per tonne of rice), mean 2022–2024. North provinces dominate, driven by irrigated single-crop systems (SF_w = 1.0) combined with moderate yields. * Province with rainfed override (SF_w = 0.52 applied despite Central/North administrative classification). † Small-area South provinces (<3 kha); intensity estimates unreliable due to sampling noise.

Int. Rank	Province	Region	SF_w	Yield (t ha^{-1})	CH_4 (Gg)	Int. (kg t^{-1})	Prod. Rank
1	Phayao	North	1.00	3.07	15.9	49.7	35
2	Nan	North	1.00	3.19	8.4	48.0	47
3	Lampang	North	1.00	3.23	7.7	46.4	48
4	Lopburi	Central	1.00	3.36	26.1	44.9	24
5	Phetchabun	North	1.00	3.45	29.2	44.4	22
6	Chiang Rai	North	1.00	3.55	38.3	42.6	13
7	Sukhothai	North	1.00	3.57	38.9	42.2	12
8	Phrae	North	1.00	3.64	7.6	41.9	46
9	Prachuap KK[b]	Central	1.00	3.61	0.6	41.6	62
10	Chiang Mai	North	1.00	3.67	15.5	41.4	30
11	Krabi†	South	0.52	1.95	0.1	40.9	73
12	Nakhon Sawan	Central	1.00	3.74	72.4	40.5	1
13	Phitsanulok	North	1.00	3.75	49.9	40.2	6
14	Surat Thani†	South	0.52	1.99	0.2	40.0	68
15	Phichit	North	1.00	3.84	55.7	39.4	4
16	Sa Kaeo*	Central	0.52	2.03	9.5	39.3	44
17	Uttaradit	North	1.00	3.83	21.9	39.2	25
18	Bueng Kan	Northeast	0.52	2.05	5.9	38.8	50
19	Kamphaeng Phet	North	1.00	3.87	31.6	38.4	16
20	Uthai Thani	Central	1.00	4.01	15.5	37.9	28

[b] Prachuap Khiri Khan (abbreviated).

Table 6. Complete provincial rice CH_4 inventory, mean annual 2022–2024, IPCC 2006 Tier 1 (76 named administrative units). Sorted by CH_4 emissions (descending). † rainfed override province. Area and production are two-season totals per province. One unattributed OAE South-region aggregate entry (73 kha; 5.75 Gg CH_4) is included in regional totals (Table A1) but excluded here; Phuket province not reported by OAE (negligible rice cultivation).

#	Province	Region	SF_w	Area (kha)	Prod. (Mt)	CH_4 (Gg)	Int. (kg t^{-1})
1	Nakhon Sawan	Central	1.00	478	1.789	72.44	40.5
2	Phichit	North	1.00	368	1.412	55.66	39.4
3	Ubon Ratchathani	Northeast	0.52	662	1.519	52.71	34.7
4	Phitsanulok	North	1.00	331	1.241	49.92	40.2
5	Suphan Buri	Central	1.00	331	1.476	49.59	33.6
6	Nakhon Ratchasima	Northeast	0.52	564	1.368	44.87	32.8
7	Surin	Northeast	0.52	491	1.177	39.10	33.2
8	Roi Et	Northeast	0.52	490	1.121	38.92	34.7

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Table 6 – continued

#	Province	Region	SF_w	Area (kha)	Prod. (Mt)	CH_4 (Gg)	Int. (kg t^{-1})
9	Sukhothai	North	1.00	258	0.922	38.90	42.2
10	Chiang Rai	North	1.00	253	0.900	38.29	42.6
11	Si Sa Ket	Northeast	0.52	477	1.090	37.97	34.9
12	Buri Ram	Northeast	0.52	459	1.066	36.55	34.3
13	Ayutthaya[a]	Central	1.00	235	1.027	35.08	34.2
14	Kamphaeng Phet	North	1.00	210	0.813	31.57	38.4
15	Khon Kaen	Northeast	0.52	376	0.833	29.91	35.9
16	Chai Nat	Central	1.00	197	0.780	29.59	37.9
17	Phetchabun	North	1.00	191	0.659	29.24	44.4
18	Maha Sarakham	Northeast	0.52	347	0.831	27.58	33.2
19	Sakon Nakhon	Northeast	0.52	343	0.752	27.29	36.3
20	Lopburi	Central	1.00	173	0.582	26.11	44.9
21	Udon Thani	Northeast	0.52	318	0.732	25.35	34.6
22	Chachoengsao	Central	1.00	150	0.609	22.55	37.0
23	Uttaradit	North	1.00	146	0.558	21.89	39.2
24	Chaiyaphum	Northeast	0.52	271	0.675	21.58	32.0
25	Kalasin	Northeast	0.52	270	0.707	21.36	30.2
26	Nakhon Phanom	Northeast	0.52	236	0.530	18.78	35.5
27	Yasothon	Northeast	0.52	213	0.517	16.94	32.7
28	Phayao	North	1.00	104	0.319	15.87	49.7
29	Chiang Mai	North	1.00	102	0.375	15.51	41.4
30	Uthai Thani	Central	1.00	102	0.408	15.46	37.9
31	Ang Thong	Central	1.00	88	0.378	13.23	35.0
32	Sing Buri	Central	1.00	85	0.370	12.79	34.6
33	Kanchanaburi	Central	1.00	85	0.341	12.77	37.5
34	Pathum Thani	Central	1.00	80	0.360	11.94	33.2
35	Nakhon Pathom	Central	1.00	77	0.366	11.41	31.2
36	Saraburi	Central	1.00	75	0.305	11.29	37.0
37	Sa Kaeo	Central	0.52[†]	120	0.243	9.53	39.3
38	Phetchaburi	Central	1.00	63	0.279	9.52	34.1
39	Nong Bua Lamphu	Northeast	0.52	114	0.262	9.06	34.6
40	Amnat Charoen	Northeast	0.52	113	0.262	9.04	31.3
41	Ratchaburi	Central	1.00	59	0.256	8.76	34.2
42	Nan	North	1.00	55	0.176	8.43	48.0
43	Nong Khai	Northeast	0.52	102	0.251	8.10	32.3
44	Lampang	North	1.00	51	0.163	7.72	46.4
45	Phrae	North	1.00	50	0.182	7.64	41.9
46	Prachin Buri	Central	0.52[†]	87	0.247	6.86	27.8
47	Nakhon Nayok	Central	0.52[†]	82	0.302	6.46	21.4
48	Bueng Kan	Northeast	0.52	74	0.153	5.92	38.8

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Table 6 – continued

#	Province	Region	SF_w	Area (kha)	Prod. (Mt)	CH_4 (Gg)	Int. (kg t^{-1})
49	Mukdahan	Northeast	0.52	73	0.186	5.79	31.2
50	Loei	Northeast	0.52	65	0.145	5.21	35.9
51	Tak	North	0.52†	60	0.153	4.78	31.2
52	Nonthaburi	Central	1.00	26	0.113	3.82	33.7
53	Bangkok	Central	1.00	25	0.104	3.70	35.7
54	Mae Hong Son	North	0.52†	34	0.085	2.70	31.7
55	Songkhla	South	0.52	29	0.100	2.33	23.3
56	Nakhon Si Thammarat	South	0.52	26	0.078	2.03	26.1
57	Lamphun	North	1.00	12	0.045	1.81	37.8
58	Phatthalung	South	0.52	21	0.063	1.67	26.6
59	Chon Buri	Central	0.52†	15	0.051	1.17	23.2
60	Pattani	South	0.52	13	0.038	1.03	27.2
61	Samut Prakan	Central	1.00	5	0.023	0.74	32.7
62	Prachuap KK[b]	Central	1.00	4	0.014	0.59	41.6
63	Narathiwat	South	0.52	4	0.013	0.34	26.9
64	Yala	South	0.52	3	0.006	0.23	37.3
65	Surat Thani	South	0.52	3	0.005	0.21	40.0
66	Satun	South	0.52	2	0.006	0.19	32.6
67	Trat	Central	0.52†	2	0.007	0.18	28.0
68	Rayong	Central	0.52†	2	0.006	0.15	24.4
69	Samut Sakhon	Central	1.00	1	0.004	0.15	34.4
70	Chanthaburi	Central	0.52†	2	0.004	0.14	36.2
71	Trang	South	0.52	1	0.004	0.12	33.5
72	Krabi	South	0.52	1	0.002	0.08	40.9
73	Samut Songkhram	Central	1.00	1	0.002	0.08	34.1
74	Chumphon	South	0.52	<1	0.001	0.03	29.8
75	Phang Nga	South	0.52	<1	0.001	0.02	37.0
76	Ranong	South	0.52	<1	0.000	0.01	34.1

[a] Phra Nakhon Si Ayutthaya. [b] Prachuap Khiri Khan.

Intensity values for provinces with <3 kha harvested area (ranks 64–76) are unreliable due to small sample sizes.

Source: OAE (Office of Agricultural Economics, 2024); IPCC 2006 (Eggleston et al., 2006); authors' calculation.

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